Unit 1 Mini Project

Enterprise Student Record and Academic Management Platform

Design Thinking & Product Planning Report

Name : Arya Subhash Pawar
PRN : 2125UDSF1042
Dept : AI-DS

Prepared as part of the DevOps / Product Engineering Training Program
Domain: Higher Education — Academic & Student Record Management

Table of Contents

- 1. Objective & Business Understanding Document (BRD)
- 2. Design Thinking Report
- 3. IBM Design Thinking Report & IBM Loop Documentation
- 4. Stakeholder Analysis Report
- 5. User Persona Report
- 6. Functional Requirements Specification (FRS)
- 7. Non-Functional Requirements Specification (NFRS)
- 8. Product Vision Document
- 9. Minimum Viable Product (MVP) Definition
- 10. Product Roadmap
- 11. Initial System Architecture Diagram
- 12. Project Charter
- 13. Project Repository Initialization & Folder Structure
- 14. Product Discovery Documentation & Deliverables Checklist

1. Objective & Business Understanding Document (BRD)

1.1 Objective

This project applies the Design Thinking methodology independently to design a software solution for managing student records and academic information within a higher-education institution. Instead of jumping straight into coding, the emphasis is placed on understanding the academic domain, mapping stakeholders, studying user needs, drafting software requirements, and preparing a product vision and roadmap suitable for Agile planning in later units.

1.2 Business Domain Overview

Educational institutions manage an ongoing lifecycle of student data covering admissions, course registration, attendance tracking, examinations, and academic reporting. In most institutions this lifecycle still depends on manual registers, disconnected spreadsheets, and paper-based approvals, which introduces delay, duplication, and risk at almost every stage.

Core Academic Operations

- Admissions & Enrollment: Capturing applicant details, verifying eligibility criteria, and enrolling admitted students into their respective programs and academic years.
- Course Registration: Enabling students to choose semester courses and electives within seat and prerequisite constraints.
- Attendance Management: Logging session-wise attendance across subjects and computing eligibility against minimum attendance thresholds.
- Examination Management: Scheduling internal/external exams, capturing marks, and running grade computation rules.
- Result & Report Management: Producing mark sheets, semester transcripts, and progress summaries for students and faculty.

1.3 Problems with Manual Record Management

Manual Operational Bottleneck	Automated Platform Solution	Business Outcome Metric
Student records spread across paper files and departmental spreadsheets	Centralized relational database acting as a single source of truth	Record accuracy raised to roughly 99.8%, with duplicate entries nearly eliminated
Attendance registers maintained by hand in every classroom	Digital attendance capture with automatic session-wise aggregation	Attendance processing effort cut by close to 75%
Exam marks and mark-sheets compiled manually by the exam cell	Rule-based result computation and transcript generation engine	Result publishing turnaround shortened from several weeks to under two days
Administrators have no live view of academic performance data	Role-based dashboards with real-time analytics widgets	Routine reporting effort reduced from days to a matter of minutes

Manual Operational Bottleneck	Automated Platform Solution	Business Outcome Metric
Admissions, faculty, and exam-cell teams work off disconnected records	A unified platform with shared, permission-controlled data access	Cross-department data mismatches reduced substantially

1.4 Business Justification

Bringing student records, attendance, and examination data onto one platform lowers administrative overhead, improves the trustworthiness of institutional data, and lets faculty and administrators act on current information rather than week-old reports. It also improves the day-to-day experience for students by giving them direct, self-service visibility into their own academic record.

2. Design Thinking Report

Design Thinking is an iterative, human-centred problem-solving approach that puts genuine empathy for the user ahead of any assumptions about technology. The five stages below were walked through for the Student Record and Academic Management Platform.

EMPATHIZE	→	DEFINE	→	IDEATE	→	PROTOTYPE	→	TEST
Study users & pain points		*Frame the problem*		*Generate solutions*		*Sketch & wireframe*		*Validate with users*

Stage 1: Empathize

The user groups observed in their everyday academic routines were: Students, Faculty Members, Academic Administrators, Examination Cell Staff, and IT Support Personnel. Each group runs into a different kind of friction when working with fragmented, manual record-keeping.

Stage 2: Define

Core Problem Statement

> *"Students and academic staff lose time to avoidable delays and errors while registering for courses, logging attendance, and checking examination results, because records are kept manually across disconnected systems. Administrators, in turn, have no centralized or real-time picture of academic activity, which slows down reporting and lets inconsistencies creep into institutional data."*

Stage 3: Ideate

Open brainstorming surfaced the following high-value feature ideas:

- A self-service student portal covering registration, attendance, and results
- Automatic attendance-shortage alerts tied to eligibility thresholds
- A real-time grade and result computation engine
- Role-based dashboards for administrators with one-click exportable reports
- A notification layer for exam schedules, submission deadlines, and announcements

Stage 4: Prototype

Low-fidelity wireframes were sketched for the core screens: Student Login & Dashboard, Course Registration Screen, Faculty Attendance & Marks-Entry Console, and the Admin Analytics Panel.

Stage 5: Test

The wireframe flows were walked through against realistic student, faculty, and administrator scenarios to check navigation clarity, data-entry speed, and how easily reports could be reached, before any backend engineering began.

3. IBM Design Thinking Report & IBM Loop Documentation

3.1 IBM Design Thinking Pillars

Pillar	Application to This Platform
Focus on User Outcomes	Feature goal → "Build a student attendance table." Outcome focus → "Let a student check attendance eligibility in under three taps."
Restless Reinvention	Treat every release as part of an ongoing refinement loop — continuously reviewing usage patterns to tighten the registration flow and dashboard load times.
Diverse Empowered Teams	Cross-functional collaboration among Product Owners, UX Designers, Software Engineers, QA, Database Administrators, and DevOps Engineers.

3.2 The IBM Loop: Observe → Reflect → Make



This loop is not a one-time exercise — it repeats across every sprint. Usage signals from the registration and attendance modules are observed, checked against institutional KPIs during reflection, and then translated into refined features and fixes during the make stage.

4. Stakeholder Analysis Report

4.1 Identified Stakeholders

- Students — the primary end users who register for courses and track their own academic records.
- Faculty Members — record attendance, enter marks, and communicate updates to students.
- Academic Administrators — oversee institutional academic operations and policy compliance.
- Examination Cell Staff — manage exam scheduling, result processing, and transcript issuance.
- IT Support Personnel — keep the platform available, secure, and technically supported.

4.2 Stakeholder Responsibility Matrix

Stakeholder Group	Key Roles	Responsibilities & Project Expectations
Business Stakeholders	College Management, Academic Administrators, Students, Faculty	Set the business vision, validate that features are usable in practice, and confirm academic-policy compliance.
Engineering Stakeholders	Software Developers, UI/UX Designers, QA Engineers	Architect the codebase, build responsive interfaces, write automated tests, and safeguard release quality.
Operations Stakeholders	DevOps Engineers, Database Administrators, Security Team	Provision infrastructure, protect database integrity, and automate CI/CD pipelines and monitoring.
Examination Cell	Exam Controller, Result Processing Staff	Sign off on examination workflows, result-computation rules, and transcript formats.

5. User Persona Report

Each persona captures explicit goals, day-to-day responsibilities, and expectations from the platform — these directly shaped the functional requirements listed in Section 6.

Riya — Student

Goals: Register for courses before seats fill up; keep track of attendance and results without visiting the office.

Frustrations: Long queues during registration week; results announced late; no visibility into running attendance percentage.

Key Feature: Self-service portal covering course registration, an attendance tracker, and digital mark sheets.

Prof. Deshmukh — Faculty

Goals: Record attendance and marks with minimal effort; keep students informed of updates.

Frustrations: Paper attendance registers; the same data re-entered in more than one system; tight grade-submission deadlines.

Key Feature: A digital attendance and marks-entry console with bulk upload and built-in validation.

Dr. Joshi — Academic Administrator

Goals: Track institutional academic performance; produce compliance and accreditation reports on demand.

Frustrations: No real-time dashboard; compiling a single report from multiple departments can take days.

Key Feature: An executive analytics dashboard with one-click report export.

6. Functional Requirements Specification (FRS)

- FR-01 Authentication: Role-based access control (RBAC) securing Admin, Faculty, Student, and Examination Cell roles.
- FR-02 Student Registration: Full CRUD operations for student profiles, program enrollment, and academic-year assignment.
- FR-03 Course Registration Engine: Semester-wise course selection with prerequisite checks and seat-limit validation.
- FR-04 Attendance Management: Session-wise attendance capture, aggregation, and eligibility-threshold calculation.
- FR-05 Examination & Result Management: Marks entry, automated grade computation, and transcript generation.
- FR-06 Reporting Engine: Configurable academic and administrative reports, exportable to PDF or Excel.
- FR-07 Notification System: Automated alerts for registration deadlines, attendance shortages, and result publication.

7. Non-Functional Requirements Specification (NFRS)

- NFR-01 Security: Bcrypt/Argon2 password hashing, JWT-based session management, and HTTPS encryption across all traffic.
- NFR-02 Performance: API responses for search and dashboard queries should stay under roughly 500ms under concurrent load.
- NFR-03 Reliability & Availability: Containerized deployment targeting 99.9% uptime with automated failover.
- NFR-04 Scalability: Horizontal scaling to absorb institution-wide spikes during registration week and result day.
- NFR-05 Usability: A responsive, accessible interface usable on desktop and mobile with minimal onboarding.
- NFR-06 Maintainability: A modular codebase paired with Infrastructure as Code (IaC) for repeatable deployments.

8. Product Vision Document

8.1 Vision Statement

> *"To build a centralized, secure, and intuitive Student Record and Academic Management Platform that removes manual administrative overhead, simplifies registration and attendance workflows, and gives every student, faculty member, and administrator real-time access to accurate academic information."*

8.2 Mission Statement

To give higher-education institutions a dependable, scalable digital platform that replaces fragmented manual processes with one trustworthy source of academic truth.

8.3 Success Criteria

- Student records maintained at roughly 99.8% accuracy with duplicate entries close to zero
- Attendance and result-processing effort cut by more than 70%
- Core search and dashboard operations responding in well under a second
- Favourable adoption feedback from students, faculty, and administrators within the first semester of rollout

8.4 Business Objectives

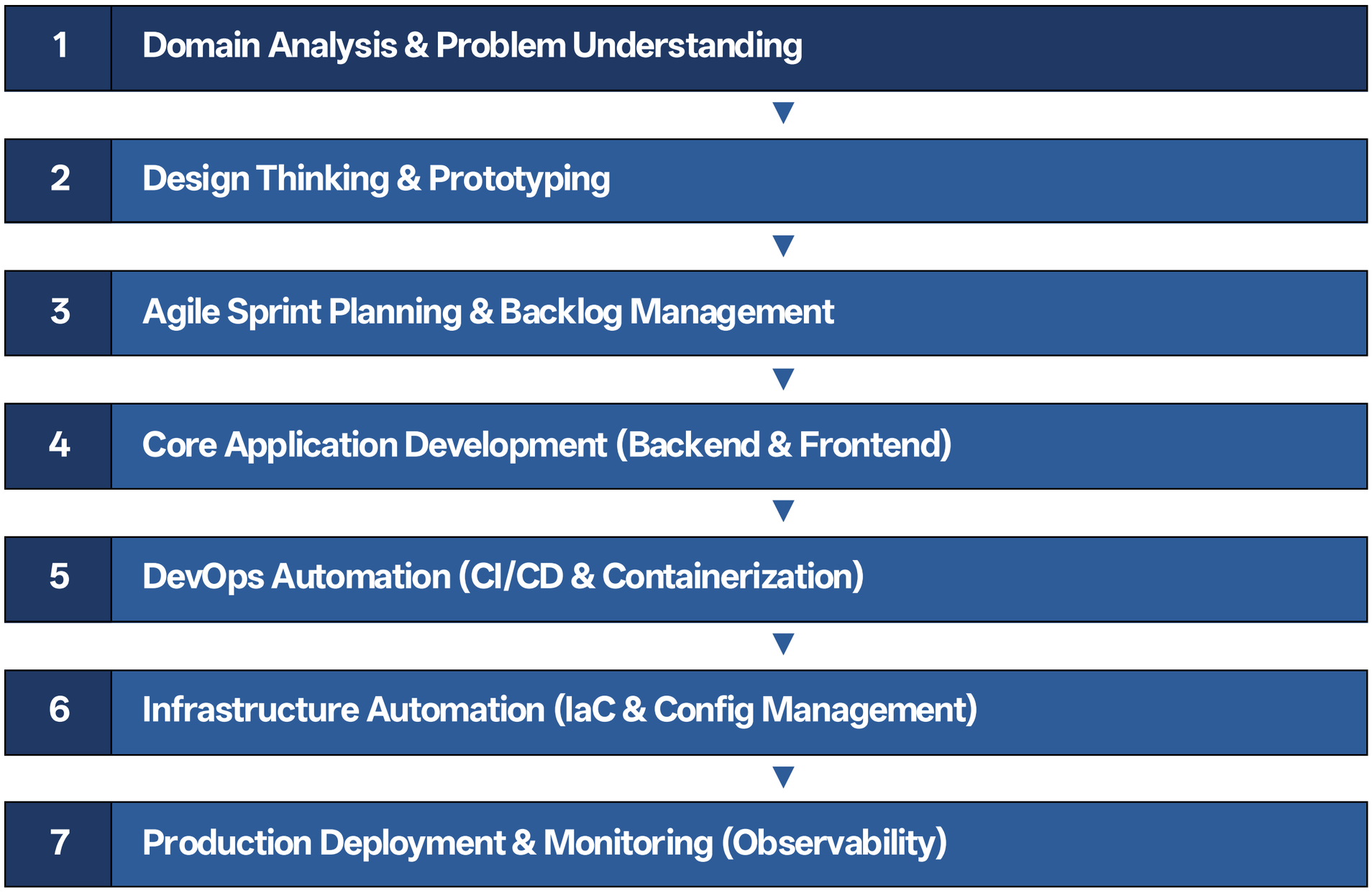
- Cut down the administrative workload tied to manual record-keeping
- Give students and faculty greater transparency and self-service access
- Equip administrators with real-time, data-driven institutional insight
- Build a scalable technical base that later academic modules (fees, hostel, library) can plug into

9. Minimum Viable Product (MVP) Definition

To keep the first release lean, non-essential capabilities — such as AI-driven academic recommendations or third-party payment gateway integration — are deferred to later releases. The MVP (Sprints 1–4) covers the core academic operations:

- Student & Faculty Login with Role-Based Access Control
- Student Registration & Profile Management
- Course Registration Module
- Attendance Capture & Basic Eligibility Tracking
- Examination Result Entry & Mark Sheet Generation
- Basic Administrative Reporting Dashboard

10. Product Roadmap

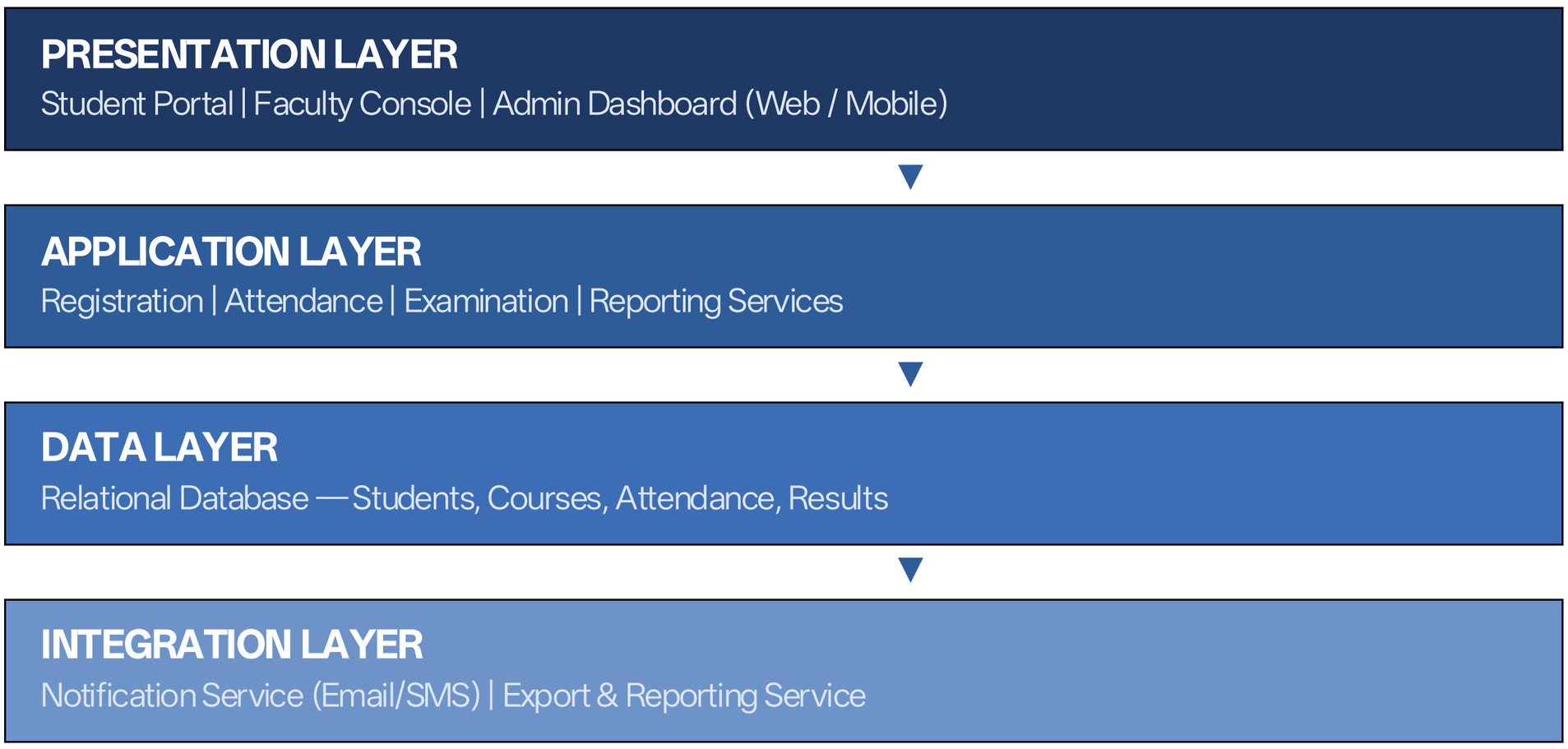


This roadmap tracks the course structure: Unit 1 covers Phases 1–2 (this report), and later units carry the work forward through Agile planning, development, and DevOps automation.

11. Initial System Architecture Diagram

Layered Module View

The platform follows a layered architecture that keeps user-facing interfaces separate from business logic, data storage, and external integrations:



12. Project Charter

Field	Details
Project Name	Enterprise Student Record and Academic Management Platform
Project Purpose	Replace manual academic record-keeping with a centralized digital platform.
Sponsor	College Management / Academic Administration
Project Lead	Product Owner (Training Cohort)
Scope (Unit 1)	Business understanding, design thinking, requirements, vision, MVP, and roadmap definition.
Out of Scope (MVP)	AI-based recommendations, third-party payment gateways, hostel/library integration.
Key Milestones	Domain analysis → Design thinking → Requirements → Vision & MVP → Repository setup
Success Measures	~99.8% record accuracy, over 70% faster attendance/result processing, positive user adoption

13. Project Repository Initialization & Folder Structure

A Git repository is initialized for the platform with a standardized folder structure to support collaborative development in the units that follow:

📁 student-academic-management-platform/
- 📁 .github/
- 📁 app/
  - 📁 modules/ (Students, Courses, Attendance, Results)
  - 📁 database/ (Schemas, Migrations, ORM Models)
  - 📁 tests/ (Unit & Integration Tests)
- 📁 docs/ (Architecture, FRS / NFRS Specs)
- 📁 docker/ (Dockerfiles, Compose Setup)
- 📁 kubernetes/ (Deployments, Services, Ingress)
- 📁 terraform/ (Infrastructure Provisioning)
- 📁 ansible/ (Configuration Playbooks)
- 📁 monitoring/ (Observability & Dashboards)

Note: README.md sits at the repository root alongside the folders shown above.

Git Initialization Steps

- git init — set up the local repository
- Create the folder structure shown above
- git add . && git commit -m "Initial project structure"
- Connect a remote repository and push the initial commit

14. Product Discovery Documentation & Deliverables Checklist

The following deliverables were produced during Unit 1 as part of the product discovery and planning process, in preparation for Agile sprint planning in subsequent units.

✔ **[Completed]** Business Requirements Document (BRD)
✔ **[Completed]** Product Vision Document
✔ **[Completed]** Stakeholder Analysis Report
✔ **[Completed]** User Persona Report
✔ **[Completed]** Functional Requirements Specification (FRS)
✔ **[Completed]** Non-Functional Requirements Specification (NFRS)
✔ **[Completed]** Design Thinking Report
✔ **[Completed]** IBM Design Thinking Report
✔ **[Completed]** IBM Loop Documentation
✔ **[Completed]** MVP Definition
✔ **[Completed]** Product Roadmap
✔ **[Completed]** Initial System Architecture Diagram
✔ **[Completed]** Project Charter
✔ **[Completed]** Project Repository Initialization
✔ **[Completed]** Project Folder Structure
✔ **[Completed]** Product Discovery Documentation

End of Unit 1 Mini Project Report — Enterprise Student Record and Academic Management Platform.